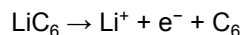
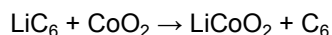


Some models of rechargeable hearing aids use lithium-ion batteries composed of a carbon (graphite) anode and a cobalt oxide-based cathode. Reaction 1 shows the half-reaction that occurs at the anode during battery discharge, and Reaction 2 shows the balanced net reaction.



Reaction 1



Reaction 2

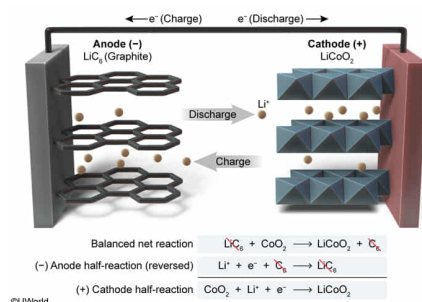
When the hearing aid is operating, what half-reaction occurs at the cathode?

- ✓ ☒ A. $\text{CoO}_2 + \text{Li}^+ + \text{e}^- \rightarrow \text{LiCoO}_2$ [69%]
☐ B. $\text{LiCoO}_2 \rightarrow \text{CoO}_2 + \text{Li}^+ + \text{e}^-$ [9%]
☐ C. $\text{CoO}_2 + \text{Li}^+ \rightarrow \text{LiCoO}_2 + \text{e}^-$ [9%]
☐ D. $\text{LiCoO}_2 + \text{e}^- \rightarrow \text{CoO}_2 + \text{Li}^+$ [11%]

Correct

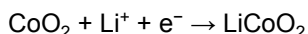
70% answered correctly

Explanation:



When a battery discharges, the half-reaction at the anode gives up electrons (**oxidation**), which are transferred to the cathode (reduction) to complete the other half-reaction. An **electron balance** must be present between the half-reactions to achieve the **balanced net reaction** from the **sum of the two half-reactions**: Electrons present as reactants in one half-reaction should appear as products in the other half-reaction. This may require **multiplication of one or both half-reactions** by a constant to ensure that all **electrons cancel**. **Identical species** appearing as reactants in one half-reaction and as products in the other cancel and are **not part of the net reaction** expression.

Based on these principles, an unknown half-reaction can be found by adding the balanced net reaction to the *inverse* of the known half-reaction. Species in the inverse known half-reaction that do not participate in the discharge (LiC_6 and C_6) will cancel with the balanced net reaction, leaving only those species present in the other half-reaction occurring during discharge:



(Choice B) This half-reaction is the *reverse* of the half-reaction that occurs at the cathode during discharge. It occurs only when the battery is being *charged*.

(Choice C) This half-reaction has the correct chemical species as products and reactants; however, reduction occurs at the cathode. The electron should be included with the reactants.

(Choice D) Although this half-reaction correctly shows an electron on the reactant side (reduction), Li^+ should also be on the reactants side of the half-reaction because it receives the electron.

Educational objective:

The balanced net reaction for an electrochemical cell may be found by taking the sum of the half-reactions, written to be consistent with the direction of electron flow within the cell. Half-reactions may be scaled stoichiometrically to achieve

an electron balance. Species matching in type and number on both sides of the summation are not included in the net reaction.

Time spent:0 sec

Subject:
General Chemistry

QID:401443

System:
Solutions and Electrochemistry